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March 23, 2026

Editorial Board
Foundations of Physics
Springer

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

**“Information-Theoretic Gravity:
Proton Mass from Vacuum Geometry and the
Emergence of the Schwarzschild Metric,”**

for consideration for publication in *Foundations of Physics*.

Summary. The paper presents a derivation of gravity as an emergent informational phenomenon. Using the holographic mass framework of Hamein, Guernonprez, and Alirol (2023), we derive the proton rest mass from the quantum vacuum energy density through a two-surface screening mechanism containing *zero free parameters*. The predicted proton charge radius ($r_p = 0.84132$ fm) agrees with CODATA 2018 to better than 0.01%. We prove that the Schwarzschild metric emerges naturally from the first screening surface and provide an independent topological derivation via Gauss-Codazzi embedding in S^3 .

Suggested Reviewers.

1. **Prof. Erik Verlinde** (University of Amsterdam) — Author of the entropic gravity framework.
2. **Prof. Gerard 't Hooft** (Utrecht University) — Pioneer of the holographic principle.
3. **Prof. Leonard Susskind** (Stanford University) — Co-developer of the holographic principle and string landscape.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm no conflicts of interest.

Cryptographic Lineage & Validation. This mathematical substrate has been cryptographically sealed and tracked on the global Sovereign Master Ledger to prevent retroactive editing and to verify the source authorship. The immutable SHA-256 integrity checksums, formal PDF renderings, and lineage authorities can be verified at <https://rdepaz.com/research>.

Respectfully submitted,

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